

Activity: Recon and Defense (Network Security I)

Part II. Reconnaissance

Q1. Notice the open ports. Does anything look suspicious, i.e., some ports that you are not aware of that are open on the VM or on your notebook?

- VM: There are 2 ports opened which are 22:ssh and 80:http.

Hosts		Services		Nmap Output		Ports / Hosts	Topology	Host Details	Scans
OS	Host				Port	Protocol	State	Service	Version
	localhost (127.0.0.1)				22	tcp	open	ssh	OpenSSH 8.9p1 Ubuntu
	192.168.64.6				80	tcp	open	http	Apache httpd 2.4.52 ((Ubuntu))

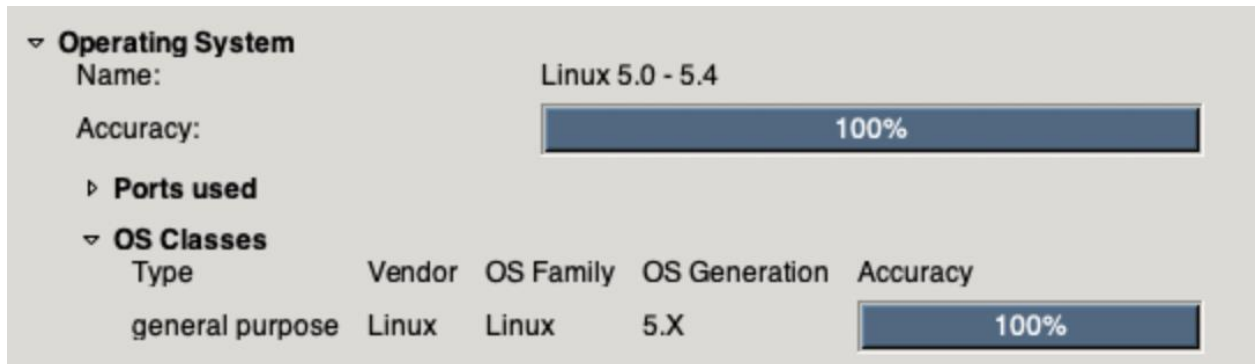
- Localhost: There are 12 ports opened and newly initialized one port 4000 which belongs to 'nmap'.

Hosts		Services		Nmap Output		Ports / Hosts	Topology	Host Details	Scans
OS	Host				Port	Protocol	State	Service	Version
	localhost (127.0.0.1)				25	tcp	open	smtp	Postfix smtpd
	192.168.64.6				53	tcp	open	domain	
					88	tcp	open	kerberos-sec	Heimdal Kerberos (serve
					445	tcp	open	microsoft-ds	
					587	tcp	open	smtp	Postfix smtpd
					3306	tcp	open	mysql	MySQL 8.0.30
					4000	tcp	filtered	remoteanything	
					5000	tcp	open	rtsp	AirTunes rtspd 620.8.2
					5432	tcp	open	postgresql	PostgreSQL DB 14.1 - 14
					5900	tcp	open	vnc	Apple remote desktop vnc
					7000	tcp	open	rtsp	AirTunes rtspd 620.8.2
					8021	tcp	open	tcpwrapped	
					49154	tcp	open	bandwidth-test	MikroTik bandwidth-test s

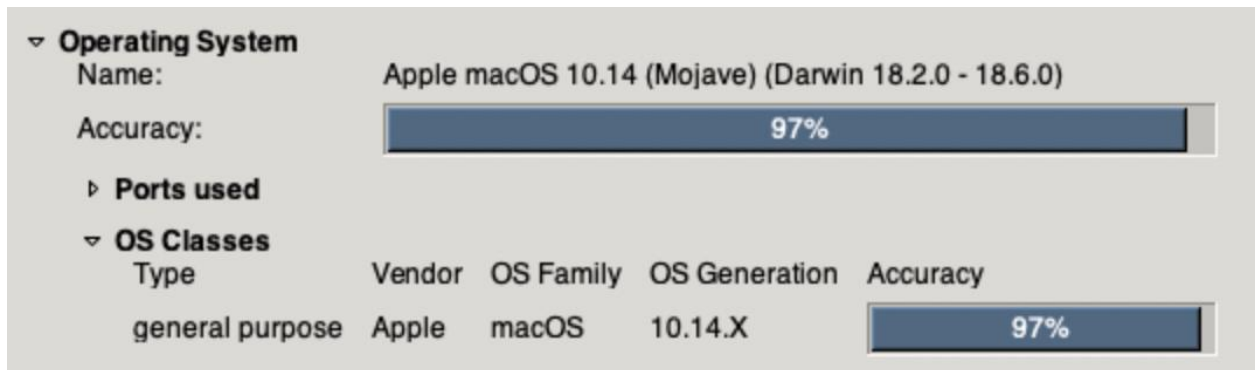
Q2. Look at the information provided by nmap about your OS's. Is the information correct? Why is it or why is it not correct?

- The OS's information was correct, but not detailed.

VM:



Localhost:



Q3. What do you think about the information you can get using nmap? Scary?

- From the 'nmap', we can acknowledge all the opened ports for available services on target, so does a hacker! This is also a sort of environment sniffing to acquire more information of the target before starting an attack.

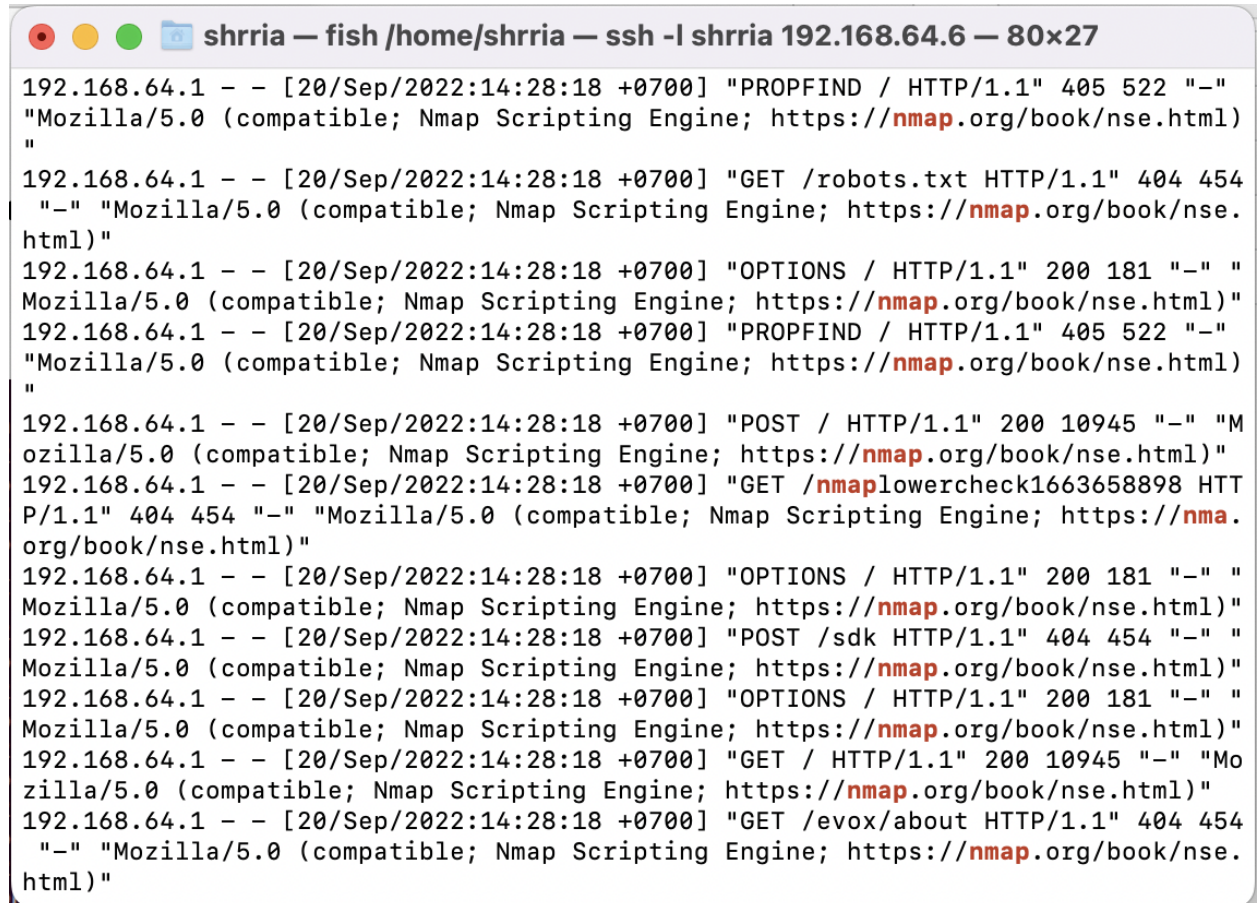
Q4. Look at the access.log file for the web server in your Linux VM. What IP addresses do you see accessing the Web server? Who owns these IP addresses?

- [IP Address: 127.0.0.1](#) was equal to 'localhost' address, in this case, belonged to the VM.
- [IP Address: 192.168.64.1](#) was assigned to my local notebook by the network of the VM.

```
shrria — vi /var/log/apache2/access.log /home/shrria — ssh -l shrria 192.1...
127.0.0.1 - - [20/Sep/2022:13:29:09 +0700] "GET / HTTP/1.1" 200 3460 "-" "Mozilla/5.0 (X11; Linux aarch64; rv:104.0) Gecko/20100101 Firefox/104.0"
127.0.0.1 - - [20/Sep/2022:13:29:09 +0700] "GET /icons/ubuntu-logo.png HTTP/1.1" 200 3607 "http://localhost/" "Mozilla/5.0 (X11; Linux aarch64; rv:104.0) Gecko/20100101 Firefox/104.0"
127.0.0.1 - - [20/Sep/2022:13:29:09 +0700] "GET /favicon.ico HTTP/1.1" 404 487 "http://localhost/" "Mozilla/5.0 (X11; Linux aarch64; rv:104.0) Gecko/20100101 Firefox/104.0"
127.0.0.1 - - [20/Sep/2022:13:30:56 +0700] "GET / HTTP/1.1" 200 3460 "-" "Mozilla/5.0 (X11; Linux aarch64; rv:104.0) Gecko/20100101 Firefox/104.0"
127.0.0.1 - - [20/Sep/2022:13:35:12 +0700] "GET / HTTP/1.1" 200 3460 "-" "Mozilla/5.0 (X11; Linux aarch64; rv:104.0) Gecko/20100101 Firefox/104.0"
127.0.0.1 - - [20/Sep/2022:13:36:31 +0700] "GET / HTTP/1.1" 200 3460 "-" "Mozilla/5.0 (X11; Linux aarch64; rv:104.0) Gecko/20100101 Firefox/104.0"
192.168.64.1 - - [20/Sep/2022:13:38:03 +0700] "GET / HTTP/1.1" 200 3460 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/15.6.1 Safari/605.1.15"
192.168.64.1 - - [20/Sep/2022:13:38:03 +0700] "GET /icons/ubuntu-logo.png HTTP/1.1" 200 3607 "http://192.168.64.6/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/15.6.1 Safari/605.1.15"
192.168.64.1 - - [20/Sep/2022:13:38:03 +0700] "GET /favicon.ico HTTP/1.1" 404 491 "http://192.168.64.6/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/15.6.1 Safari/605.1.15"
1,80 Top
```

Q5. Find the nmap scan in the Web server log. Copy the lines from the log file that were created because of the nmap scan.

- `grep nmap /var/log/apache2/access.log`



```
shrria — fish /home/shrria — ssh -l shrria 192.168.64.6 — 80x27
192.168.64.1 - - [20/Sep/2022:14:28:18 +0700] "PROPFIND / HTTP/1.1" 405 522 "-"
"Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)
"
192.168.64.1 - - [20/Sep/2022:14:28:18 +0700] "GET /robots.txt HTTP/1.1" 404 454
 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.
html)"
192.168.64.1 - - [20/Sep/2022:14:28:18 +0700] "OPTIONS / HTTP/1.1" 200 181 "-" "
Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
192.168.64.1 - - [20/Sep/2022:14:28:18 +0700] "PROPFIND / HTTP/1.1" 405 522 "-"
"Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)
"
192.168.64.1 - - [20/Sep/2022:14:28:18 +0700] "POST / HTTP/1.1" 200 10945 "-" "M
ozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
192.168.64.1 - - [20/Sep/2022:14:28:18 +0700] "GET /nmaplowercheck1663658898 HTT
P/1.1" 404 454 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nma
.org/book/nse.html)"
192.168.64.1 - - [20/Sep/2022:14:28:18 +0700] "OPTIONS / HTTP/1.1" 200 181 "-" "
Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
192.168.64.1 - - [20/Sep/2022:14:28:18 +0700] "POST /sdk HTTP/1.1" 404 454 "-" "
Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
192.168.64.1 - - [20/Sep/2022:14:28:18 +0700] "OPTIONS / HTTP/1.1" 200 181 "-" "
Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
192.168.64.1 - - [20/Sep/2022:14:28:18 +0700] "GET / HTTP/1.1" 200 10945 "-" "Mo
zilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
192.168.64.1 - - [20/Sep/2022:14:28:18 +0700] "GET /evox/about HTTP/1.1" 404 454
 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.
html)"
```

Part III. Defense

Q6. After you successfully install your iptable rule(s), how do the reported results from your new nmap scan compare to your previous scan before using iptables? Look to see if OS detection, port open results, etc. have changed. Something(s) have definitely changed.

- [OS information](#) was received similarly.



- [Ports](#) opened were detected as previous because we left them receiving inputs from anywhere.

OS	Host		Port	Protocol	State	Service	Version
	192.168.64.1		22	tcp	open	ssh	OpenSSH 8.9p1 Ubuntu 3 (Ubuntu Linux; protocol 2.0)
	172.20.10.3		80	tcp	open	http	Apache httpd 2.4.52 ((Ubuntu))

- Also, there was a warning message of [OSScan results](#) shown in logging this time!

Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port

Q7. Notice that nmap can still figure out you have Apache httpd running. Look at the access.log file for the web server in your Linux VM. Are the logs the same as in Part II?

- Yes. The Apache2 logs displayed just the same, even if firewall rules were changed.

```
shrria — vim /var/log/apache2/access.log /home/shrria — ssh -l shrria 172...
192.168.64.1 - - [20/Sep/2022:20:03:24 +0700] "GET / HTTP/1.0" 200 10945 "-" "-"
192.168.64.1 - - [20/Sep/2022:20:03:35 +0700] "OPTIONS / HTTP/1.1" 200 181 "-" "
Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
192.168.64.1 - - [20/Sep/2022:20:03:35 +0700] "GET / HTTP/1.1" 200 10945 "-" "Mo
zilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
192.168.64.1 - - [20/Sep/2022:20:03:35 +0700] "GET /nmaplowercheck1663679015 HTT
P/1.1" 404 454 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap
.org/book/nse.html)"
192.168.64.1 - - [20/Sep/2022:20:03:35 +0700] "GET /.git/HEAD HTTP/1.1" 404 454
 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.h
tml)"
192.168.64.1 - - [20/Sep/2022:20:03:35 +0700] "OPTIONS / HTTP/1.1" 200 181 "-" "
Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
192.168.64.1 - - [20/Sep/2022:20:03:35 +0700] "PROPFIND / HTTP/1.1" 405 522 "-"
 "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)
"
192.168.64.1 - - [20/Sep/2022:20:03:35 +0700] "POST /sdk HTTP/1.1" 404 454 "-" "
Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
192.168.64.1 - - [20/Sep/2022:20:03:35 +0700] "POST / HTTP/1.1" 200 10945 "-" "M
ozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
192.168.64.1 - - [20/Sep/2022:20:03:35 +0700] "PROPFIND / HTTP/1.1" 405 522 "-"
 "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)
"
192.168.64.1 - - [20/Sep/2022:20:03:35 +0700] "GET / HTTP/1.0" 200 10945 "-" "-"
192.168.64.1 - - [20/Sep/2022:20:03:35 +0700] "OPTIONS / HTTP/1.1" 200 181 "-" "
Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
192.168.64.1 - - [20/Sep/2022:20:03:35 +0700] "GET /robots.txt HTTP/1.1" 404 454
 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.
html)"
192.168.64.1 - - [20/Sep/2022:20:03:35 +0700] "OPTIONS / HTTP/1.1" 200 181 "-" "
Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
192.168.64.1 - - [20/Sep/2022:20:03:35 +0700] "GET /favicon.ico HTTP/1.1" 404 45
4 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse
.html)"
192.168.64.1 - - [20/Sep/2022:20:03:35 +0700] "PROPFIND / HTTP/1.1" 405 522 "-"
@@@
1,1 Top
```

Q8. Explain how you could prevent nmap from reaching the web server while still allowing legitimate clients to get service. Will a firewall be sufficient for this? Or do you need some other device? Please think critically about this.

- Actually, a firewall was not sufficient to prevent nmap from reaching the web server. The DROP action for closed ports could help delay the time of scanning by nmap to make the hackers took longer duration when searching for the opened ports.
- To secure the server from nmap, we need to adopt techniques such as 'Port Knocking', 'Honeypots and Honeynets', and 'Reactive Port Scan Detection'.

Q9. What are your firewall rules? Run *iptables -L* on your VM and enter the output here.

- Output of the command.

```
root@Shrria-Ubuntu /h/shrria# iptables -L
Chain INPUT (policy DROP)
target     prot opt source                destination          tcp dpt:http
ACCEPT     tcp  --  anywhere               anywhere             tcp dpt:ssh
ACCEPT     tcp  --  172.20.10.2            anywhere

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination
root@Shrria-Ubuntu /h/shrria#
```